LEXAN™ FR Resin 3412ECR - Asia

Polycarbonate

SABIC

PROSPECTOR®
www.ulprospector.com

Technical Data

Product Description			
LEXAN™ 3412ECR resin is a 20% glass fiber filled, 7 MFR polycarbonate, MVR of 7. Mold release. Non-chlorinated, non-brominated flame retardant, UL94 V0 and 5VA rated. Available in natural and opaque colors.			

General			
Material Status	• Commercial: Active		
UL Yellow Card [1]	• E207780-228394 • E207780-102101507		
Search for UL Yellow Card	• SABIC		
Availability	• Asia Pacific		
Uses	• Appliances • Automotive Exterior Parts • Construction Applications • Electrical Parts • Electrical/Electronic Applications	• Electronic Displays • Industrial Applications • Lighting Applications • Material Handling • Medical/Healthcare Applications	• Military/Defense Applications • Rail Applications • Water Management
Also Available In	• Europe	• Latin America	• North America

Physical	Nominal Value (English)	Nominal Value (SI)	Test Method
Density / Specific Gravity			
--	1.30	1.30 g/cm³	ASTM D792
--	1.36 g/cm³	1.36 g/cm³	ISO 1183
Melt Mass-Flow Rate (MFR) (300°C/1.2 kg)	7.0 g/10 min	7.0 g/10 min	ASTM D1238
Melt Volume-Flow Rate (MVR) (300°C/1.2 kg)	7.0 cm³/10min	7.0 cm³/10min	ISO 1133
Molding Shrinkage			Internal Method
Across Flow : 0.126 in (3.20 mm)	0.20 to 0.50 %	0.20 to 0.50 %	
Flow : 0.126 in (3.20 mm)	0.20 to 0.50 %	0.20 to 0.50 %	
Water Absorption			ISO 62
Saturation, 73°F (23°C)	0.29 %	0.29 %	
Equilibrium, 73°F (23°C), 50% RH	0.12 %	0.12 %	

Mechanical	Nominal Value (English)	Nominal Value (SI)	Test Method
Tensile Modulus			
-- [3]	798000 psi	5500 MPa	ASTM D638
--	870000 psi	6000 MPa	ISO 527-1/1
Tensile Strength			
Yield [4]	13100 psi	90.0 MPa	ASTM D638
Yield	13800 psi	95.0 MPa	ISO 527-2/5
Break [4]	12600 psi	87.0 MPa	ASTM D638
Break	13100 psi	90.0 MPa	ISO 527-2/5
Tensile Elongation			
Yield [4]	3.1 %	3.1 %	ASTM D638
Yield	2.8 %	2.8 %	ISO 527-2/5
Break	3.2 %	3.2 %	ISO 527-2/5
Flexural Modulus			
1.97 in (50.0 mm) Span [5]	725000 psi	5000 MPa	ASTM D790
-- [6]	798000 psi	5500 MPa	ISO 178
Flexural Stress			
-- [6, 7]	20300 psi	140 MPa	ISO 178
Yield, 1.97 in (50.0 mm) Span [5]	22600 psi	156 MPa	ASTM D790

Impact	Nominal Value (English)	Nominal Value (SI)	Test Method
Charpy Notched Impact Strength [8]			ISO 179/1eA
-22°F (-30°C)	2.4 ft·lb/in²	5.0 kJ/m²	
73°F (23°C)	2.9 ft·lb/in²	6.0 kJ/m²	

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Impact	Nominal Value (English)	Nominal Value (SI)	Test Method
Charpy Unnotched Impact Strength [8]			ISO 179/1eU
-22°F (-30°C)	19 ft·lb/in²	40 kJ/m²	
73°F (23°C)	19 ft·lb/in²	40 kJ/m²	
Notched Izod Impact			
-22°F (-30°C)	2.0 ft·lb/in	110 J/m	ASTM D256
73°F (23°C)	2.1 ft·lb/in	110 J/m	ASTM D256
-22°F (-30°C) [9]	2.9 ft·lb/in²	6.0 kJ/m²	ISO 180/1A
73°F (23°C) [9]	3.3 ft·lb/in²	7.0 kJ/m²	ISO 180/1A
Unnotched Izod Impact Strength [9]			ISO 180/1U
-22°F (-30°C)	17 ft·lb/in²	35 kJ/m²	
73°F (23°C)	17 ft·lb/in²	35 kJ/m²	
Instrumented Dart Impact			ASTM D3763
73°F (23°C), Total Energy	177 in·lb	20.0 J	

Thermal	Nominal Value (English)	Nominal Value (SI)	Test Method
Deflection Temperature Under Load			
66 psi (0.45 MPa), Unannealed, 0.157 in (4.00 mm), 2.52 in (64.0 mm) Span [10]	286 °F	141 °C	ISO 75-2/Bf
264 psi (1.8 MPa), Unannealed, 0.126 in (3.20 mm)	286 °F	141 °C	ASTM D648
264 psi (1.8 MPa), Unannealed, 0.157 in (4.00 mm), 2.52 in (64.0 mm) Span [10]	277 °F	136 °C	ISO 75-2/Af
Vicat Softening Temperature			
--	297 °F	147 °C	ASTM D1525 [11]
--	295 °F	146 °C	ISO 306/B120
--	293 °F	145 °C	ISO 306/B50
Ball Pressure Test			IEC 60695-10-2
253 to 261°F (123 to 127°C)	Pass	Pass	
CLTE			
Flow : -40 to 104°F (-40 to 40°C)	1.7E-5 in/in/°F	3.0E-5 cm/cm/°C	ASTM E831
Flow : 73 to 176°F (23 to 80°C)	1.7E-5 in/in/°F	3.0E-5 cm/cm/°C	ISO 11359-2
Transverse : -40 to 104°F (-40 to 40°C)	3.9E-5 in/in/°F	7.0E-5 cm/cm/°C	ASTM E831
Transverse : 73 to 176°F (23 to 80°C)	3.9E-5 in/in/°F	7.0E-5 cm/cm/°C	ISO 11359-2
RTI Elec	266 °F	130 °C	UL 746B
RTI Imp	266 °F	130 °C	UL 746B
RTI Str	266 °F	130 °C	UL 746B

Electrical	Nominal Value (English)	Nominal Value (SI)	Test Method
Surface Resistivity	> 1.0E+15 ohms	> 1.0E+15 ohms	IEC 60093
Volume Resistivity	> 1.0E+15 ohms·cm	> 1.0E+15 ohms·cm	IEC 60093
Electric Strength			IEC 60243-1
0.0787 in (2.00 mm), in Oil	860 V/mil	34 kV/mm	
Relative Permittivity			IEC 60250
50 Hz	3.30	3.30	
60 Hz	3.30	3.30	
1 MHz	3.30	3.30	
Dissipation Factor			IEC 60250
50 Hz	0.020	0.020	
60 Hz	0.020	0.020	
1 MHz	0.010	0.010	
Arc Resistance [12]	PLC 7	PLC 7	ASTM D495
Comparative Tracking Index (CTI)	PLC 3	PLC 3	UL 746A
Comparative Tracking Index	175 V	175 V	IEC 60112
High Amp Arc Ignition (HAI) [13]	PLC 3	PLC 3	UL 746A

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Electrical	Nominal Value (English)	Nominal Value (SI)	Test Method
High Voltage Arc Resistance to Ignition (HVAR)	PLC 3	PLC 3	UL 746A
Hot-wire Ignition (HWI)	PLC 0	PLC 0	UL 746A

Flammability	Nominal Value (English)	Nominal Value (SI)	Test Method
Flame Rating			UL 94
0.06 in (1.5 mm)	V-0	V-0	
0.12 in (3.0 mm)	5VA	5VA	
Glow Wire Flammability Index			IEC 60695-2-12
0.04 in (1.0 mm)	1760 °F	960 °C	
Glow Wire Ignition Temperature			IEC 60695-2-13
0.04 in (1.0 mm)	1520 °F	825 °C	
Oxygen Index	40 %	40 %	ISO 4589-2

Injection	Nominal Value (English)	Nominal Value (SI)
Drying Temperature	248 °F	120 °C
Drying Time	3.0 to 4.0 hr	3.0 to 4.0 hr
Suggested Max Moisture	0.020 %	0.020 %
Suggested Shot Size	40 to 60 %	40 to 60 %
Rear Temperature	509 to 554 °F	265 to 290 °C
Middle Temperature	527 to 572 °F	275 to 300 °C
Front Temperature	554 to 590 °F	290 to 310 °C
Nozzle Temperature	536 to 581 °F	280 to 305 °C
Processing (Melt) Temp	554 to 590 °F	290 to 310 °C
Mold Temperature	158 to 203 °F	70 to 95 °C
Back Pressure	43.5 to 102 psi	0.300 to 0.700 MPa
Screw Speed	40 to 70 rpm	40 to 70 rpm
Vent Depth	9.8E-4 to 3.0E-3 in	0.025 to 0.076 mm

Injection Notes

- Drying Time (Cumulative): 48 hr

Notes

[1] A UL Yellow Card contains UL-verified flammability and electrical characteristics. UL Prospector continually works to link Yellow Cards to individual plastic materials in Prospector, however this list may not include all of the appropriate links. It is important that you verify the association between these Yellow Cards and the plastic material found in Prospector. For a complete listing of Yellow Cards, visit the UL Yellow Card Search.

[2] Typical properties: these are not to be construed as specifications.

[3] 0.20 in/min (5.0 mm/min)

[4] Type I, 0.20 in/min (5.0 mm/min)

[5] 0.051 in/min (1.3 mm/min)

[6] 0.079 in/min (2.0 mm/min)

[7] at Yield

[8] 80*10*3 sp=62mm

[9] 80*10*3 mm

[10] 80*10*4 mm

[11] Rate A (50°C/h), Loading 2 (50 N)

[12] Tungsten Electrode

[13] Surface

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Where to Buy

Supplier

SABIC
Web: http://www.sabic.com/

Distributor

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Telephone: +86-20-82582555
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Availability: Asia Pacific

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